

VIRGINIA HIGHLANDS, VA, USA

WMO: 724058

Lat: 36.683N

Lon: 82.033W

Elev: 2070

StdP: 13.63

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 53818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	17.7	0.6	6.1	19.8	5.3	7.8	23.2	20.5	41.6	18.7	39.6	4.4	260	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	89.7	71.5	87.6	71.1	84.7	70.4	75.5	83.7	74.4	82.0	73.3	80.3	5.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	133.0	78.0	72.4	129.5	77.4	71.7	126.6	76.9	40.6	83.9	39.3	82.3	38.4	80.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.6	15.5	12.7	DB	5.2	91.8	6.1	2.4	0.8	93.5	-2.8	94.9	-6.2	96.2	-10.6	97.9
			WB	4.2	77.2	5.8	1.3	0.0	78.1	-3.4	78.9	-6.7	79.6	-10.9	80.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	36.2	40.0	47.3	56.4	64.7	71.8	74.4	74.0	68.7	57.7	46.9	39.7	(6)
	DBStd	15.79	11.16	10.13	9.81	8.33	6.85	4.65	3.70	3.95	6.00	8.39	8.94	9.73	(7)
	HDD50	1489	447	304	169	37	2	0	0	0	0	29	160	341	(8)
	HDD65	4130	894	701	550	271	91	9	1	1	34	250	544	784	(9)
	CDD50	3887	18	24	85	230	459	653	758	744	560	267	67	22	(10)
	CDD65	1050	0	0	1	14	83	212	293	280	144	22	1	0	(11)
	CDH74	7906	0	4	19	176	679	1593	2120	2048	1072	184	10	1	(12)
	CDH80	2304	0	0	1	21	126	474	671	672	316	22	1	0	(13)
(14) Wind	WSAvg	4.6	5.7	5.8	6.1	5.8	4.4	3.7	3.3	3.1	3.9	4.2	4.7	5.2	(14)
(15) Precipitation	PrecAvg	45.7	4.1	3.8	4.7	3.8	4.3	4.1	5.2	3.9	2.6	2.2	3.0	3.9	(15)
	PrecMax	59.1	6.7	7.8	8.0	8.2	7.9	7.0	8.1	6.7	5.3	5.0	6.2	7.3	(16)
	PrecMin	31.9	2.1	1.5	2.6	1.6	1.6	1.7	1.9	1.3	0.5	0.2	0.7	1.5	(17)
	PrecStd	6.6	1.2	1.5	1.7	1.5	1.6	1.2	1.4	1.3	1.4	1.2	1.4	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	72.1	76.7	82.3	86.2	90.4	91.2	91.5	90.1	82.4	74.7	69.4	(19)
		MCWB	57.8	58.1	58.7	63.3	67.3	71.1	73.0	73.0	67.7	67.6	61.2	60.2	(20)
	2%	DB	61.5	64.7	72.2	79.1	83.4	87.9	88.3	89.6	86.7	79.1	69.8	63.0	(21)
		MCWB	54.0	54.3	57.4	61.8	66.8	71.0	72.3	72.4	68.5	64.4	57.7	54.6	(22)
	5%	DB	57.0	60.1	67.0	75.1	81.1	84.9	86.1	86.5	83.8	75.1	65.6	58.2	(23)
		MCWB	52.0	51.7	54.6	59.8	66.2	70.1	72.1	71.7	68.5	62.1	56.3	53.1	(24)
	10%	DB	53.6	55.2	63.3	72.0	78.6	82.3	83.9	83.8	81.0	72.1	62.0	54.6	(25)
		MCWB	49.7	48.6	52.7	58.5	65.4	69.3	71.7	70.7	67.7	61.8	53.8	50.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.8	61.7	61.6	67.5	72.0	76.2	77.1	77.1	74.4	71.7	64.2	61.4	(27)
		MCDB	63.6	67.8	70.9	78.4	80.5	84.7	85.4	85.8	81.9	77.3	68.8	65.1	(28)
	2%	WB	56.6	57.2	59.3	64.5	70.1	74.3	75.6	75.3	72.5	68.7	61.3	57.6	(29)
		MCDB	60.0	62.5	68.6	74.1	78.6	82.4	83.6	83.8	79.8	74.7	66.9	60.6	(30)
	5%	WB	53.3	53.6	57.2	62.4	68.5	72.9	74.5	74.2	70.9	66.0	58.4	54.6	(31)
		MCDB	56.6	58.0	64.8	71.0	77.0	80.4	81.8	81.8	78.0	71.9	63.6	57.0	(32)
	10%	WB	49.4	49.8	54.9	60.4	67.1	71.4	73.4	72.9	69.4	63.2	55.1	50.9	(33)
		MCDB	52.6	54.4	60.9	68.5	75.3	79.0	79.9	79.8	76.6	68.7	60.0	54.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	18.0	20.2	21.7	20.8	19.9	18.3	19.5	20.3	21.2	19.7	16.6	(35)
		MCDBR	21.6	24.8	25.5	26.5	24.4	23.5	21.3	22.5	24.7	25.5	24.6	21.2	(36)
	5% WB	MCWBR	15.2	16.0	14.0	12.6	10.6	9.3	8.1	8.7	9.0	12.3	14.3	14.7	(37)
		MCWBR	17.7	20.4	21.7	21.8	21.0	19.4	17.9	18.7	18.9	18.8	19.7	17.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.291	0.300	0.323	0.364	0.402	0.432	0.462	0.464	0.368	0.338	0.304	0.286	(40)
		taud	2.534	2.504	2.478	2.376	2.271	2.218	2.166	2.136	2.393	2.480	2.547	2.573	(41)
	Ebn at Noon		289	299	300	291	280	270	261	258	280	280	282	284	(42)
		Edh at Noon	25	29	32	38	43	45	47	48	35	29	24	22	(43)
(44) All-Sky Solar Radiation	RadAvg		702	893	1263	1611	1780	1924	1789	1685	1440	1123	832	614	(44)
	RadStd		57	111	94	123	134	123	119	81	126	135	86	59	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+364	+157	(46)
(47)	Regional (40 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+155	(47)

Nomenclature: See separate page